

ASSIGNMENT -2

Problem statement-Array Puzzle solver

Create a java application that solves different types of puzzles using array manipulation.the system will include the variety of puzzles involving arrays such as finding missing numbers ,sorting and searching the problems ,finding the largest/smallest element,sum pair puzzles and other logical tasks based on arrays.

Code

```
import java.util.Arrays;
public class Puzzle{
    public static int missingNum(int[] arr, int n){
        int sum=0;
        int expected=(n*(n+1))/2;
        for(int i=0;i<arr.length;i++){
            sum=sum+arr[i];
        }

        int miss_no=expected-sum;
        return miss_no;
    }

    /*public static void sortNo(int[] arr){
        int temp;
        for(int i=0;i<arr.length;i++){
            for(int j=0;j<arr.length-i-1;j++){
                if(arr[j]>arr[j+1]){
                    temp=arr[j];
                    arr[j]=arr[j+1];
                    arr[j+1]=temp;
                }
            }
        }
    }*/

    public static void sort(int[] arr){
        Arrays.sort(arr);
    }

    public static int searchVal(int[] arr, int key){
        for(int i=0;i<arr.length;i++){
            if(key==arr[i]){
                return i;
            }
        }
        return -1;
    }
}
```

```

public static int largestNum(int[] arr){
    int largest=arr[0];
    for(int i=0;i<arr.length;i++){
        if(arr[i]>largest){
            largest=arr[i];
        }
    }
    return largest;
}

public static int smallestNum(int[] arr){
    int smallest=arr[0];
    for(int i=0;i<arr.length;i++){
        if(arr[i]<smallest){
            smallest=arr[i];
        }
    }
    return smallest;
}

public static void sumPair(int[] arr, int target) {
    boolean found = false;

    for (int i = 0; i < arr.length; i++) {
        for (int j = i + 1; j < arr.length; j++) {
            if (arr[i] + arr[j] == target) {
                System.out.println("Pair found: (" + arr[i] + ", " + arr[j] + ")");
                found = true;
            }
        }
    }

    if (!found) {
        System.out.println("No pair found with given sum");
    }
}

public static void main(String[] args){
    Puzzle p=new Puzzle();
    int n=5;
    int[] arr1={1,2,4,5};
    int m = p.missingNum(arr1,n);
    System.out.println("Missing Number: "+m);

    int[] arr2={4,2,3,9,7};
    System.out.println("Sorted Array: ");
    Arrays.sort(arr2);
}

```

```

        for(int i=0;i<arr2.length;i++){
            System.out.println(arr2[i]);
        }

        //int key=3;
        int index= p.searchVal(arr1,5);

        if(index!=-1){
            System.out.println("Target Element is not found in array");
        }else{
            System.out.println("Target Element is found at index : "+index );
        }

        int[] arr3={54,12,34,90,23};
        int l=p.largestNum(arr3);
        System.out.println("Largest Element: "+l);

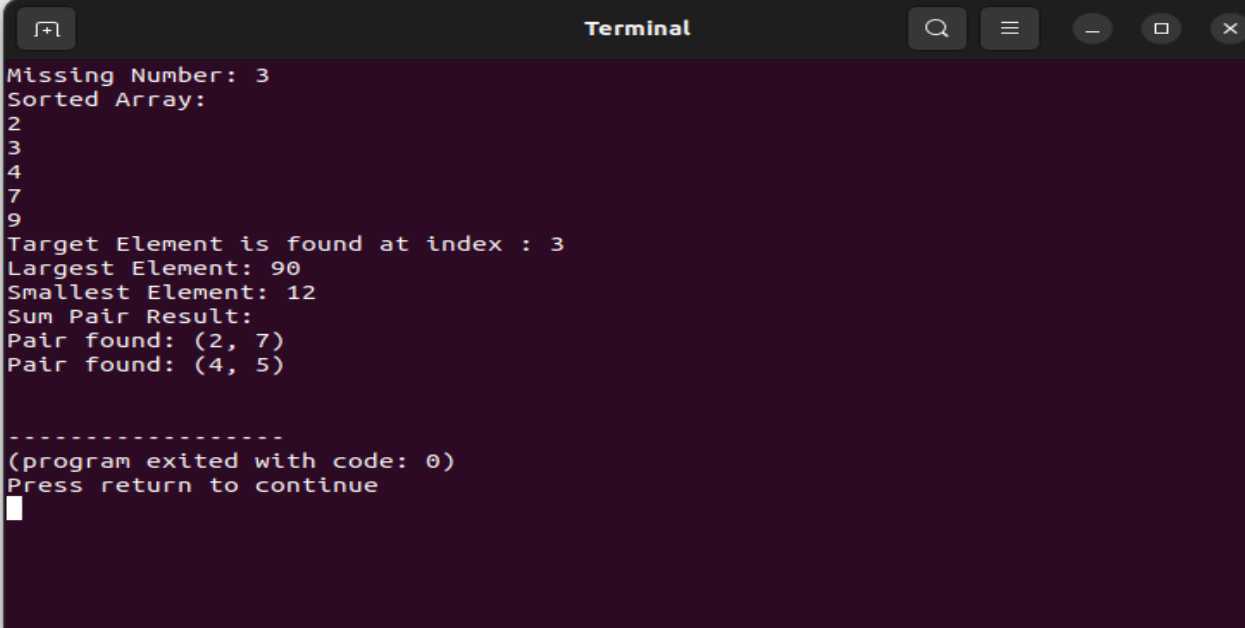
        int s=p.smallestNum(arr3);
        System.out.println("Smallest Element: "+s);

        int[] arr4 = {2, 4, 3, 5, 7};
        int target = 9;
        System.out.println("Sum Pair Result:");
        sumPair(arr4, target);

    }
}

```

Output



```

Terminal
Missing Number: 3
Sorted Array:
2
3
4
7
9
Target Element is found at index : 3
Largest Element: 90
Smallest Element: 12
Sum Pair Result:
Pair found: (2, 7)
Pair found: (4, 5)

-----
(program exited with code: 0)
Press return to continue

```

